ECE343 Spring 2025

Objective 3.4: Impedance Matching



Lesson	Topics	Reading	Notes
1	Course Overview & Introduction	Ch.1	LO 1.1-1.2
Chapter 2: Electric and Magnetic Fields			
2	Fields, Parameters, Field Quantities	Ch. 2	LO 2.1-2.2
Chapter 3: Transmission Lines			
3	Lumped Element Model	3.1-3.4	LO 3.1
4	Telegrapher's Equations	3.5-3.6	LO 3.1
5	Characteristic Impedance & Wave Propagation	3.7-3.9	LO 3.2
6	Coaxial, Reflection Coefficient, VSWR	3.10,3.12-3.14	LO 3.2-3.3
7	Terminated Transmission Lines	3.15,3.17-3.19	LO 3.4
8	Impedance Matching	3.21-3.23	LO 3.4
9	Impedance Matching	3.21-3.23	LO 3.4
10	Smith Charts	Supplemental	LO 3.5
11	Smith Charts	Supplemental	LO 3.5
12	Smith Charts	Supplemental	LO 3.5
13	Smith Charts	Supplemental	LO 3.5
14	Transients	Supplemental	LO 3.6
15	Reflection/Transmission Coefficient Lab	Lab Packet	LO 3.7
Chapter 4: Vector Analysis			
16	Vector & Coordinate System Review	Ch. 4	LO 4.1-4.4 **GR 1 due**
Chapter 5: Electrostatics			
17	Coulomb's Law	5.1-5.3	LO 5.1
18	Surface and Volume Charge	5.4	LO 5.1
19	Gauss' Law (Integral Form)	5.5-5.6	LO 5.2
20	Gauss' Law (Differential Form)	5.7-5.8	LO 5.2
21	Electric Potential	5.9-5.10, 5.15	LO 5.3

Ch 3 Learning Objectives

3.1 I can describe the origin of the transmission line equation, understand the method of developing a solution, and interpret the solution.

3.2 I can explain the physical meaning of and calculate the following quantities: propagation constant, phase constant, attenuation constant, characteristic impedance, VSWR, reflection coefficient, and transmission coefficient.

3.3 I can use the transmission line equation to calculate the voltage/current at an arbitrary point on a specified transmission line.

3.4 I can use the transmission line equation to match a load to a source using a length of transmission line.

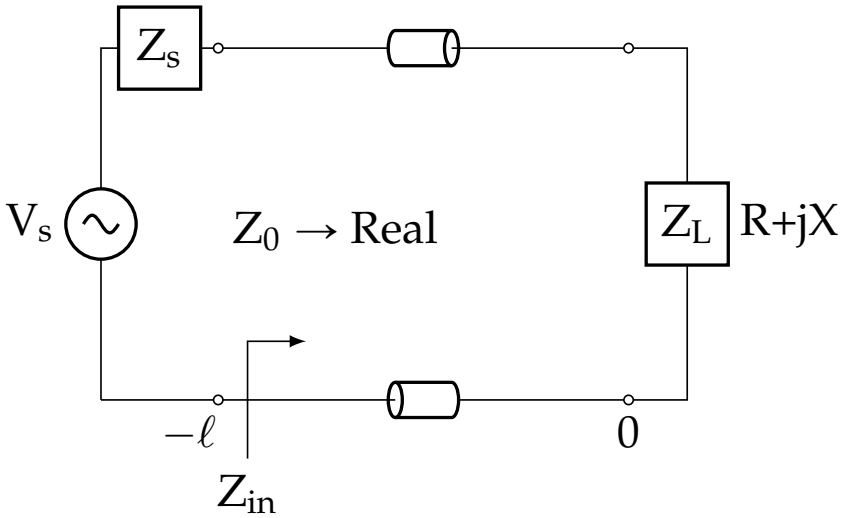
3.5 I can identify the various quantities represented on the Smith Chart.

3.6 I can use a Smith Chart to design a matching network for a given source and load, using series components, parallel components, shorted stubs, and open stubs.

3.7 I can plot the transient response of a load connected to a source through a transmission line.

3.8 I can measure the reflection coefficient across a range of frequencies and interpret the measurement using a network analyzer.

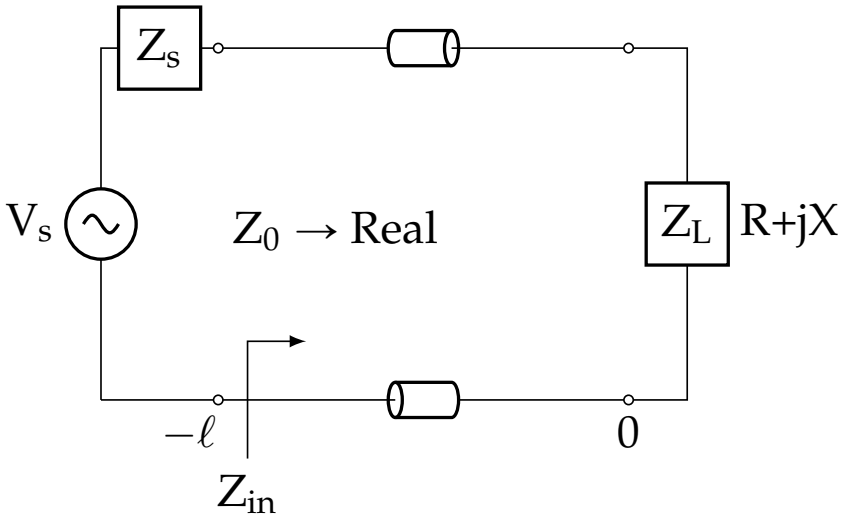
Tx Line Equation



Goal

Match Z_s to Z_{in}, so all P_s is delivered to the load.

Tx Line Equation



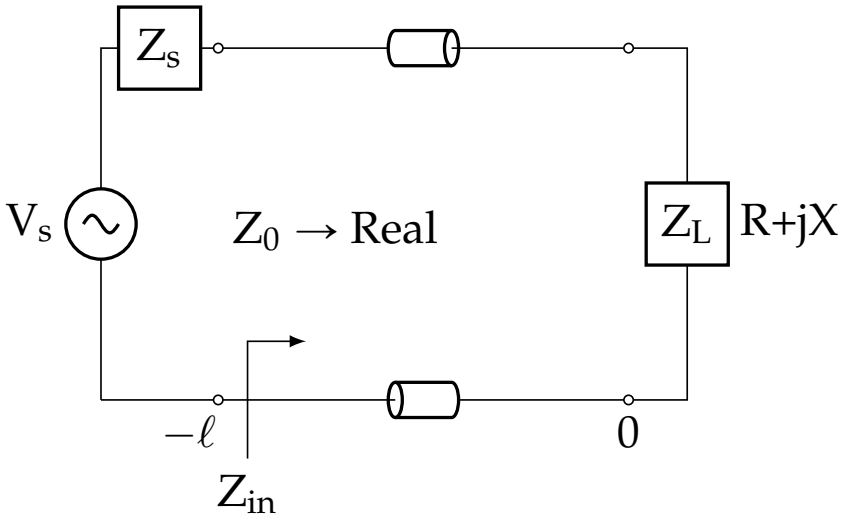
Goal

Match Z_s to Z_{in}, so all P_s is delivered to the load.

Recall the solutions to the voltage and current wave equations:

$$\widetilde{V} = V_0^+ e^{-\gamma z} + V_0^- e^{\gamma z}$$
$$\widetilde{I} = I_0^+ e^{-\gamma z} + I_0^- e^{\gamma z}$$

Tx Line Equation



Goal

Match Z_s to Z_{in}, so all P_s is delivered to the load.

Recall the solutions to the voltage and current wave equations:

$$\tilde{V} = V_0^+ e^{-\gamma z} + V_0^- e^{\gamma z}$$
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With, $\Gamma(z) = |\Gamma_L| e^{2\gamma z} = \frac{V_0^-}{V_0^+} e^{2\gamma z}$, we can write the **lossless case** as:

$$\tilde{V} = V_0^+ e^{-j\beta z} \left(1 + |\Gamma_L| e^{j2\beta z}\right)$$
$$\tilde{I} = I_0^+ e^{-j\beta z} \left(1 + |\Gamma_L| e^{j2\beta z}\right)$$

No ABS

minus sign NOT account for

Tx Line Equation

Now, we can express the **input impedance** (Z_{in}), at $z = -\ell$ as:

$$Z_{in} = \frac{\widetilde{V}}{\widetilde{I}} = \frac{V_0^+ e^{-j\beta z}\left(1 + |\Gamma_L| e^{j2\beta z}\right)}{I_0^+ e^{-j\beta z}\left(1 + |\Gamma_L| e^{j2\beta z}\right)} = Z_0 \left[\frac{1 + \Gamma_L e^{-j2\beta\ell}}{1 - \Gamma_L e^{-j2\beta\ell}}\right]$$

Tx Line Equation

Now, we can express the **input impedance** (Z_{in}), at $z = -\ell$ as:

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$$= Z_0 \left[\frac{1 + \left(\frac{Z_L - Z_0}{Z_L + Z_0}\right) e^{-j2\beta\ell}}{1 - \left(\frac{Z_L - Z_0}{Z_L + Z_0}\right) e^{-j2\beta\ell}}\right]$$

Tx Line Equation

Now, we can express the **input impedance** (Z_{in}), at $z = -\ell$ as:

$$Z_{in} = \frac{\widetilde{V}}{\widetilde{I}} = \frac{V_0^+ e^{-j\beta z}\left(1 + |\Gamma_L| e^{j2\beta z}\right)}{I_0^+ e^{-j\beta z}\left(1 + |\Gamma_L| e^{j2\beta z}\right)} = Z_0 \left[\frac{1 + \Gamma_L e^{-j2\beta\ell}}{1 - \Gamma_L e^{-j2\beta\ell}}\right]$$

$$= Z_0 \left[\frac{1 + \left(\frac{Z_L - Z_0}{Z_L + Z_0}\right) e^{-j2\beta\ell}}{1 - \left(\frac{Z_L - Z_0}{Z_L + Z_0}\right) e^{-j2\beta\ell}}\right]$$

$$= Z_0 \left[\frac{(Z_L + Z_0)\, e^{+j\beta\ell} + (Z_L - Z_0)\, e^{-j\beta\ell}}{(Z_L + Z_0)\, e^{+j\beta\ell} - (Z_L - Z_0)\, e^{-j\beta\ell}}\right]$$

Tx Line Equation

Now, we can express the **input impedance** (Z_{in}), at $z = -\ell$ as:

minus sign accounted for!

$$Z_{in} = \frac{\widetilde{V}}{\widetilde{I}} = \frac{V_0^+ e^{-j\beta z}\left(1 + |\Gamma_L| e^{j2\beta z}\right)}{I_0^+ e^{-j\beta z}\left(1 + |\Gamma_L| e^{j2\beta z}\right)} = Z_0\left[\frac{1 + \Gamma_L e^{-j2\beta\ell}}{1 - \Gamma_L e^{-j2\beta\ell}}\right]$$

$$= Z_0\left[\frac{1 + \left(\frac{Z_L - Z_0}{Z_L + Z_0}\right) e^{-j2\beta\ell}}{1 - \left(\frac{Z_L - Z_0}{Z_L + Z_0}\right) e^{-j2\beta\ell}}\right]$$

$$= Z_0\left[\frac{(Z_L + Z_0)\, e^{+j\beta\ell} + (Z_L - Z_0)\, e^{-j\beta\ell}}{(Z_L + Z_0)\, e^{+j\beta\ell} - (Z_L - Z_0)\, e^{-j\beta\ell}}\right]$$

$$= Z_0\left[\frac{Z_L 2\cos(\beta\ell) + jZ_0 2\sin(\beta\ell)}{Z_0 2\cos(\beta\ell) + jZ_L 2\sin(\beta\ell)}\right]$$

$$Z_{in} = Z_0\left[\frac{Z_L + jZ_0\tan(\beta\ell)}{Z_0 + jZ_L\tan(\beta\ell)}\right]$$

Tx Line Equation

General

$$Z_{in} = Z_0 \left[\frac{Z_L + jZ_0 \tanh(\gamma \ell)}{Z_0 + jZ_L \tanh(\gamma \ell)} \right]$$

$$Z_{in} = \frac{z_L + j\tanh(\gamma \ell)}{1 + jz_L \tanh(\gamma \ell)}$$

$$z_L = \frac{Z_L}{Z_0}$$

Lossless

$$Z_{in} = Z_0 \left[\frac{Z_L + jZ_0 \tan(\beta \ell)}{Z_0 + jZ_L \tan(\beta \ell)} \right]$$

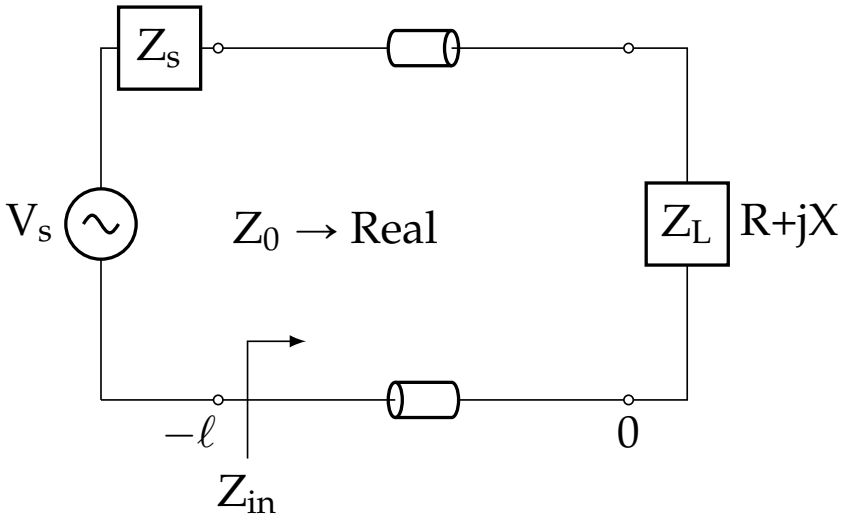
$$Z_{in} = Z_0 \left[\frac{z_L + j\tan(\beta \ell)}{1 + jz_L \tan(\beta \ell)} \right]$$

$$z_L = \frac{Z_L}{Z_0}$$

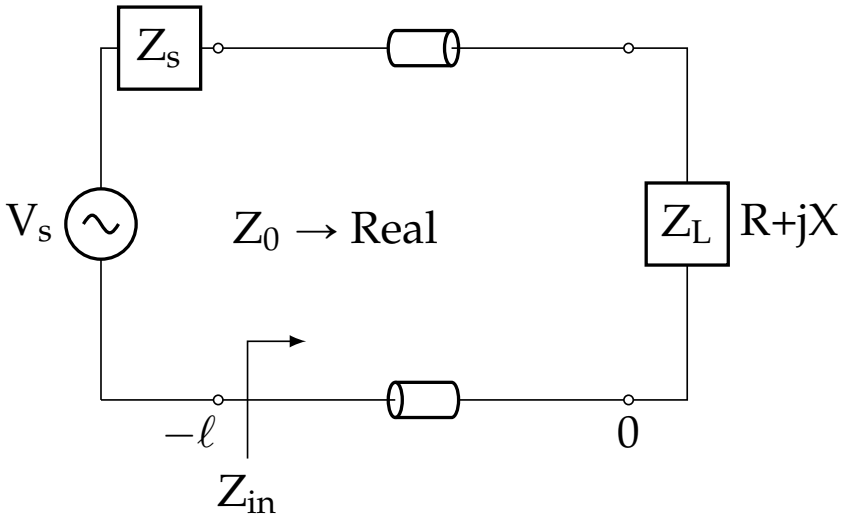
$z_L \rightarrow$ Normalized

$Z_L \rightarrow$ Unitized (Ω)

Tx Line Equation Example: $\ell = \lambda/2$

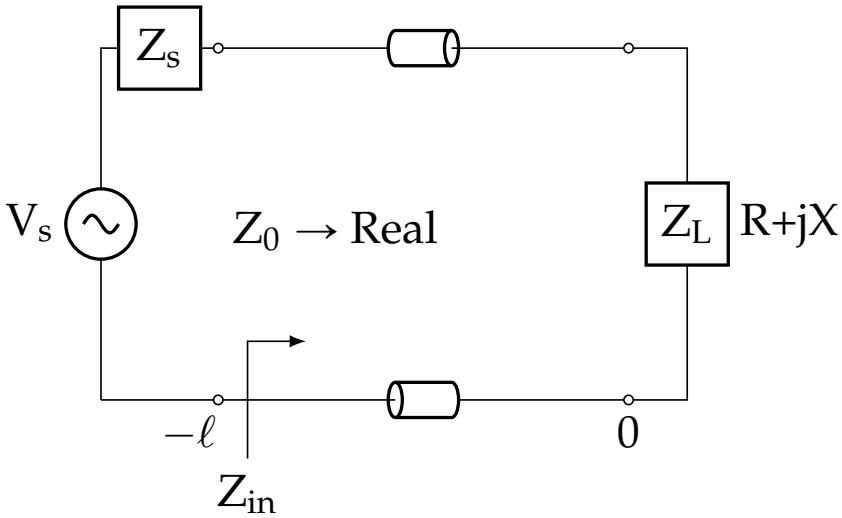


Tx Line Equation Example: $\ell = \lambda/2$



$$\beta = \frac{2\pi}{\lambda} \text{ and } \beta\ell = \pi$$

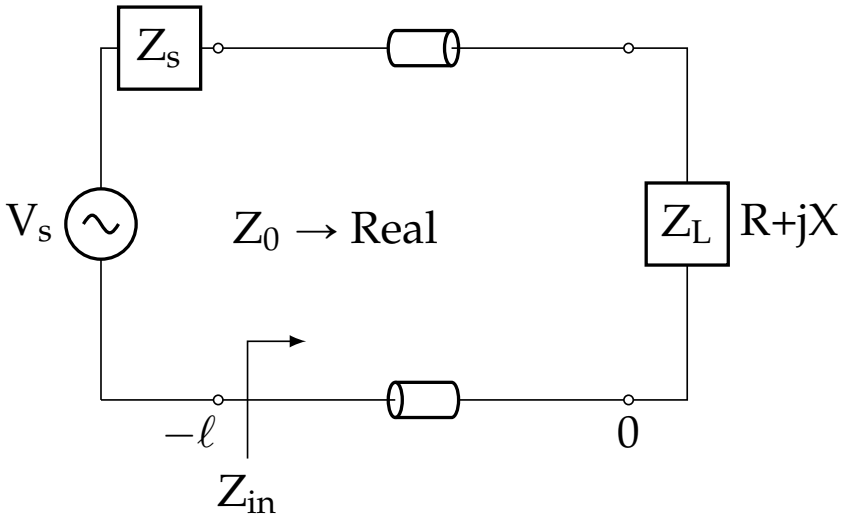
Tx Line Equation Example: $\ell = \lambda/2$



$$\beta = \frac{2\pi}{\lambda} \text{ and } \beta\ell = \pi$$

$$= \left[Z_0 \frac{z_L + j\tan(\pi)}{1 + jz_L\tan(\pi)} \right]$$

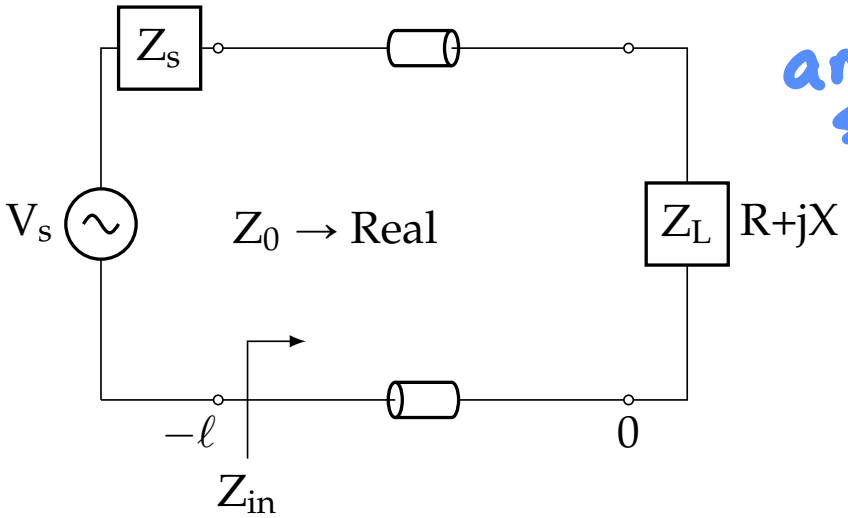
Tx Line Equation Example: $\ell = \lambda/2$



$$\beta = \frac{2\pi}{\lambda} \text{ and } \beta\ell = \pi$$
$$= \left[Z_0 \frac{z_L + j\tan(\pi)}{1 + jz_L\tan(\pi)} \right]$$
$$= \frac{z_L}{1}$$

Tx Line Equation Example: $\ell = \lambda/2$

another special case $\ell = \lambda/4$ → Quarter Wave transformer



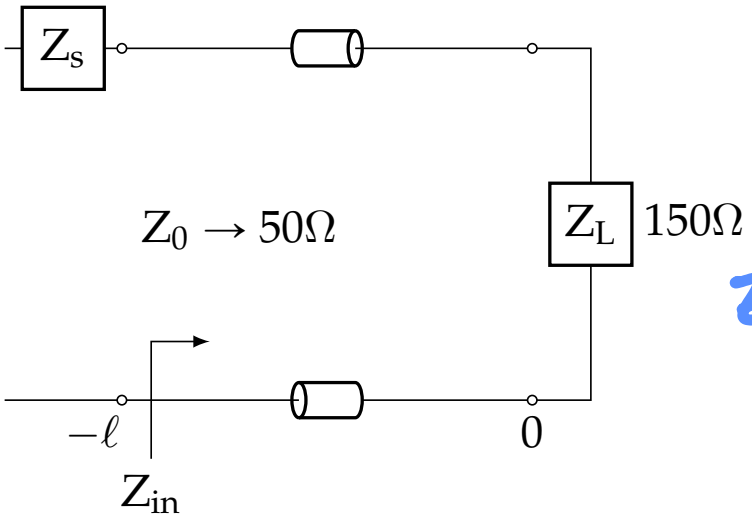
Result

A tx line with a length of $\frac{n\lambda}{2}$ is "transparent".

$$\begin{aligned}\beta &= \frac{2\pi}{\lambda} \text{ and } \beta\ell = \pi \\ &= \left[Z_0 \frac{z_L + j\tan(\pi)}{1 + jz_L\tan(\pi)}\right] \\ &= \frac{z_L}{1}\end{aligned}$$

Tx Line Equation: Example #2

Find Γ_L, VSWR, and $Z_{in}(z = -\frac{\lambda}{8})$:



1. $\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0} = \frac{z_L - 1}{z_L + 1}$

$= \frac{3-1}{3+1} = \frac{2}{4} = \frac{1}{2}$

2. VSWR

$= \frac{1 + |\Gamma_L|}{1 - |\Gamma_L|} = \frac{1 + 1/2}{1 - 1/2} = \frac{3/2}{1/2} = 3$

$z_L = \frac{150\Omega}{50\Omega} = 3$

3. $Z_{in}(z = -\frac{\lambda}{8})$

$z_{in} = \frac{Z_{in}}{Z_0}$

$z_{in} = \frac{z_L + j\tan(\beta\ell)}{1 + j z_L \tan(\beta\ell)}$

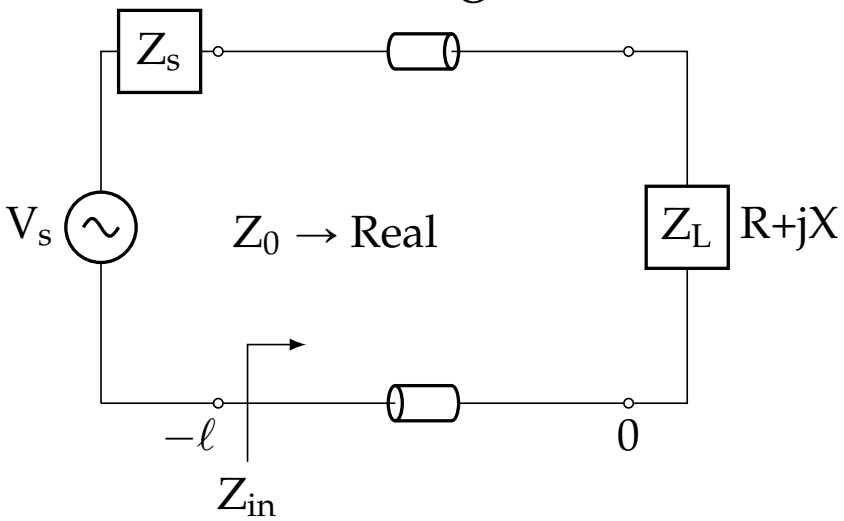
$= \frac{3 + j\tan(\frac{2\pi}{\lambda} \cdot \frac{\lambda}{8})}{1 + j3\tan(\frac{\pi}{4})}$

$= \frac{3 + j}{1 + j3} = 0.6 - j0.8$

$Z_{in} = z_{in} \cdot Z_0 = 30 - j40\Omega$

Impedance Matching

Now, back to our goal of matching load (Z_L and Tx Line) to Z_s...



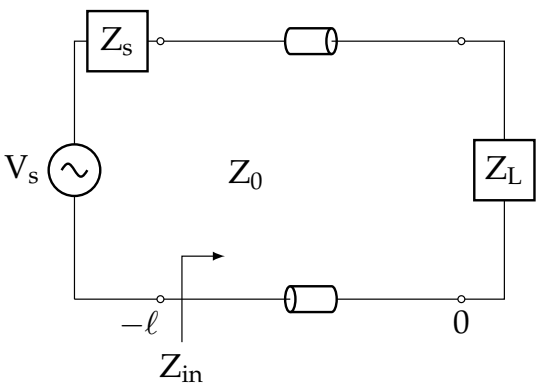
3. $Z_{in}(z = -\frac{\lambda}{8})$

1. Γ_L

2. VSWR

Impedance Matching

Now, back to our goal of matching load (Z_L and Tx Line) to Z_s...



Process

1. Normalize impedances to Z_0
2. Find reflection coefficient (Γ_L)
3. Holding the _____ of Γ_L constant, find the input impedance $z_{in} =$ _____

$$\implies |\Gamma_L| = \frac{z_{in} - 1}{z_{in} + 1}$$

4. Determine capacitance or inductance from X.

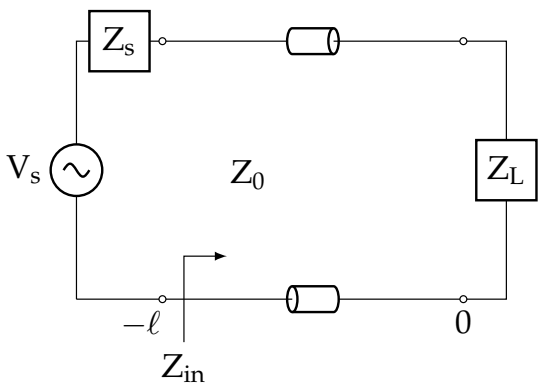
$$\implies Z_C = \frac{1}{j\omega C} \text{ or } Z_L = j\omega L$$

5. Find the length of TX line to make:

$$\implies z_{in} = 1 + jX = \frac{z_L + j\tan(\beta\ell)}{1 + jz_L\tan(\beta\ell)}$$

Impedance Matching

Now, back to our goal of matching load (Z_L and Tx Line) to Z_s...



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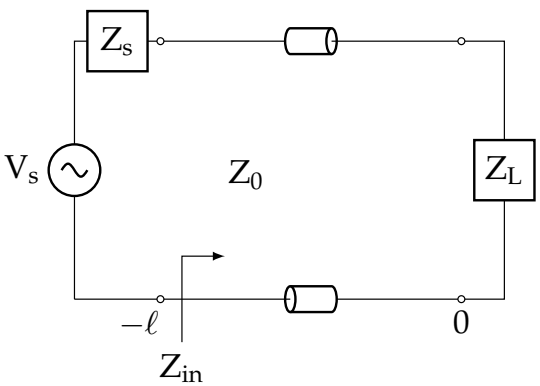
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Impedance Matching

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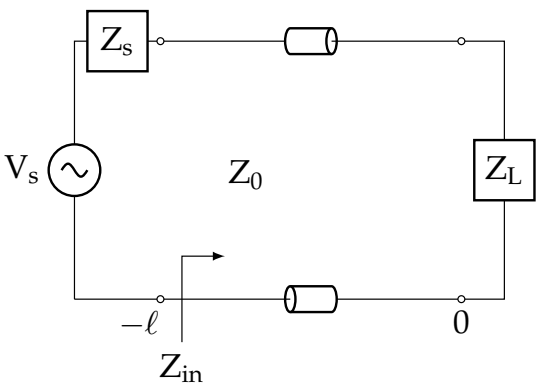
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Impedance Matching

Now, back to our goal of matching load (Z_L and Tx Line) to Z_s...



Process

1. Normalize impedances to Z_0
2. Find reflection coefficient (Γ_L)
3. Holding the _____ of Γ_L constant, find the input impedance $z_{\text{in}} =$ _____

$$\implies |\Gamma_L| = \frac{z_{\text{in}} - 1}{z_{\text{in}} + 1}$$

4. Determine capacitance or inductance from X.

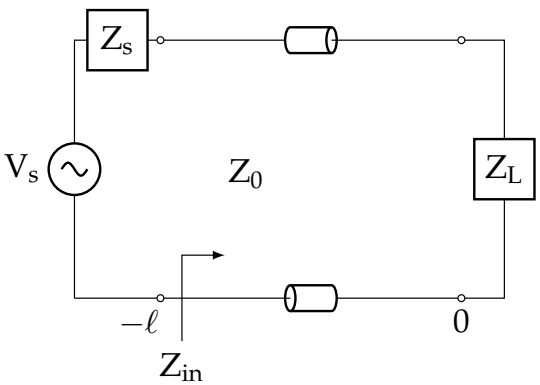
$$\implies Z_C = \frac{1}{j\omega C} \text{ or } Z_L = j\omega L$$

5. Find the length of TX line to make:

$$\implies z_{\text{in}} = 1 + jX = \frac{z_L + j\tan(\beta\ell)}{1 + jz_L\tan(\beta\ell)}$$

Impedance Matching

Now, back to our goal of matching load (Z_L and Tx Line) to Z_s...



Process

1. Normalize impedances to Z_0
2. Find reflection coefficient (Γ_L)
3. Holding the Mag. of Γ_L constant, find the input impedance $z_{in} =$ Re[z_L] $\pm jX$

$$\Longrightarrow |\Gamma_L| = \frac{z_{in} - 1}{z_{in} + 1}$$

4. Determine capacitance or inductance from X.

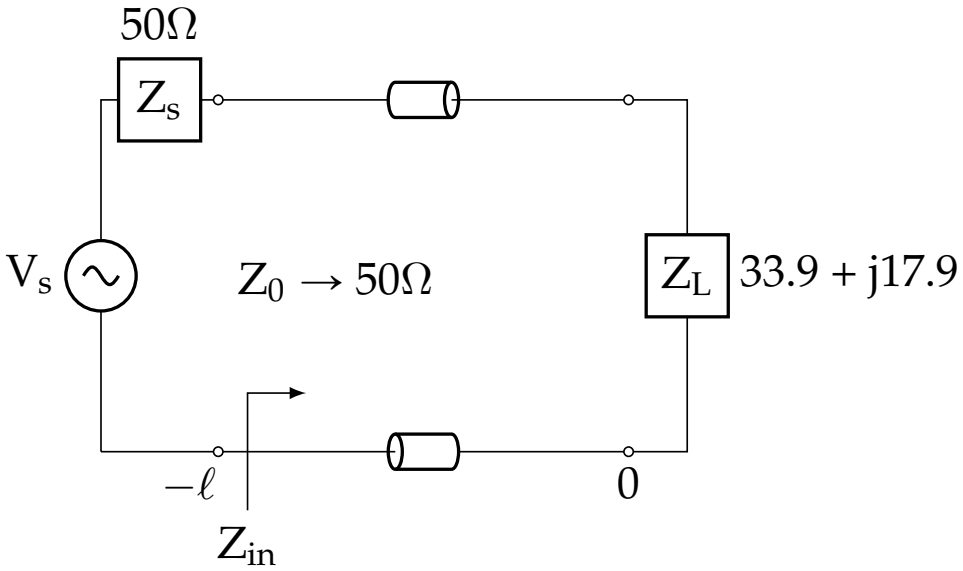
$$\Longrightarrow Z_C = \frac{1}{j\omega C} \text{ or } Z_L = j\omega L$$

5. Find the length of TX line to make:

$$\Longrightarrow z_{in} = 1 + jX = \frac{z_L + j\tan(\beta\ell)}{1 + jz_L\tan(\beta\ell)}$$

Impedance Matching: Example

Match the combined load (Z_L and TX line) to Z_s.



1. Normalize Impedances

$z_s = 1$

$z_0 = 1$

$z_L = \frac{33.9 + j17.9\,\Omega}{50\,\Omega} = 0.678 + j0.358$

2. Find reflection coefficient

$\Gamma_L = \frac{z_L - 1}{z_L + 1} = \frac{0.678 + j0.358 - 1}{0.678 + j0.358 + 1}$

$= -0.141 + j\,0.239$

$= 0.278 \angle 120.7^\circ$

Impedance Matching: Example (cont'd)

3. Find z_{in} & C or L

$$|\Gamma_{in}| = \left|\frac{z_{in}-1}{z_{in}+1}\right| = 0.278$$

$$z_{in} = \mathrm{Re}[z_s] + jX = 1 + jX$$

$$\left|\frac{1+jx-1}{1+jx+1}\right| = 0.278$$

You picked $x = \pm 0.579$

$$z_{in} = 1 + j0.579$$

$$\left[\frac{\sqrt{x^2}}{\sqrt{2^2+x^2}} = 0.278\right]^2$$

$$\frac{1}{j\omega C} = -j0.579$$

$$C = \frac{1}{2\pi f \cdot 0.579 \cdot Z_0}$$

$$\frac{x^2}{4+x^2} = 0.077$$

$$\vdots$$

$$x = \pm 0.579$$

Impedance Matching: Example (cont'd)

4. Determine Tx Line Length ℓ

$$Z_{in} = 1 + j0.579 = \frac{Z_L + j\tan(\beta\ell)}{1 + jZ_L\tan(\beta\ell)}$$

$\beta\ell = \frac{2\pi}{\lambda}\ell$

$$1 + j0.579 = \frac{0.678 + j0.358 + j\tan(\beta\ell)}{1 + j(0.678 + j0.358)(j\tan(\beta\ell))}$$

$$(1 + j0.579)(1 + j0.678\tan(\beta\ell) - 0.358\tan(\beta\ell)) = 0.678 + j0.358 + j\tan(\beta\ell)$$

$$1 + j0.678\tan(\beta\ell) - 0.358\tan(\beta\ell) + j0.579 - 0.393\tan(\beta\ell) - j0.207\tan(\beta\ell) = 0.678 + j0.358 + j\tan(\beta\ell)$$

① collect Re $\tan(\beta\ell)$; Im $\tan(\beta\ell)$

② $A\tan(\beta\ell) + jB\tan(\beta\ell) = C + jD$

$B\tan(\beta\ell) = D$

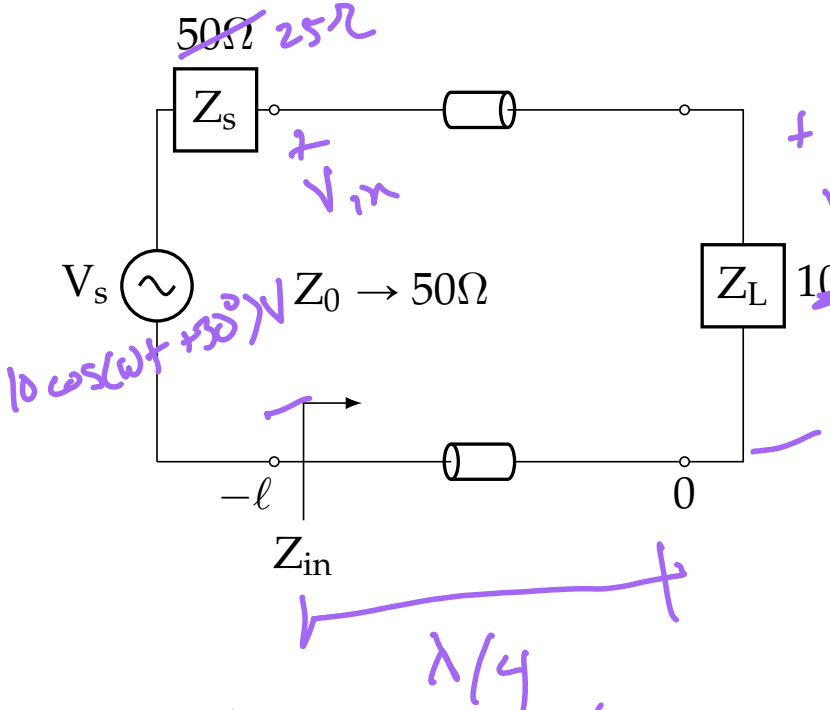
$$\beta\ell = \tan^{-1}\left(\frac{D}{B}\right) = \frac{2\pi}{\lambda}\ell \Rightarrow \boxed{\ell = \frac{\lambda\tan^{-1}\left(\frac{D}{B}\right)}{2\pi}}$$

for $Z_{in} = 1 + j0.579$

$\ell = 0.065\lambda$

Impedance Matching: Review

Match the combined load (Z_L and TX line) to Z_s at 1GHz.



Overview:

1. Find $\tilde{V}_s$
2. Find Γ_L
3. Find Z_{in}
4. Find V_{in} ← Voltage Divider
5. Find V_0^+

Find V_L

100Ω

(1) $\tilde{V}_s = 10\angle 30°$ V

(2) $\Gamma_L = \frac{100\Omega - 50\Omega}{100\Omega + 50\Omega} = \frac{1}{3}$

Aside...

$$Z_{in} = Z_0 \frac{Z_L + jZ_0 \tan\left(\frac{2\pi}{\lambda}\cdot\frac{\lambda}{4}\right)}{Z_0 + jZ_L \tan\left(\frac{\pi}{2}\right)}$$

$\tan\left(\frac{\pi}{2}\right) \longrightarrow \infty$

$$= Z_0 \frac{Z_L + jZ_0(\infty)}{Z_0 + jZ_L(\infty)} \cancel{\frac{\infty}{\infty}} = Z_0 \frac{\frac{Z_L}{\infty} + jZ_0}{\frac{Z_0}{\infty} + jZ_L} = Z_0 \frac{jZ_0}{jZ_L}$$

$$\ell = \frac{\lambda}{4} \longrightarrow Z_{in} = \frac{Z_0^2}{Z_L}$$ Quarter-Wave Transformer

$$\ell = \frac{\lambda}{2} \longrightarrow Z_{in} = Z_L$$

Now Back to our problem...



(3) $$Z_{in} = \frac{50^2}{100} = 25\Omega$$

(4) $$V_{in} = \tilde{V}_s \frac{Z_{in}}{Z_{in} + Z_s} = 10\angle 30^\circ V \left[\frac{25\Omega}{50\Omega}\right] = 5\angle 30^\circ V.$$

$$V_{in} = \tilde{V}\left(z = -\frac{\lambda}{4}\right)$$

⑤ Find V_0^+ ...

$$\tilde{V}(z) = V_0^+ \left(e^{-j\beta z} + \Gamma_L e^{j\beta z} \right)$$

$$\tilde{V}\left(z = -\frac{\lambda}{4}\right) = V_{in} = 5\angle 30^\circ = V_0^+ \left(e^{-j\frac{2\pi}{\lambda}\cdot\left(-\frac{\lambda}{4}\right)} + \frac{1}{3} e^{j\frac{2\pi}{\lambda}\left(-\frac{\lambda}{4}\right)} \right)$$

$$5\angle 30^\circ = V_0^+ \left(e^{+j\frac{\pi}{2}} + \frac{1}{3} e^{-j\pi/2} \right)$$

$$5\angle 30^\circ = V_0^+ \left(1\angle 90^\circ + \frac{1}{3}\angle -90^\circ \right)$$

$$V_0^+ = 7.5\angle -60^\circ \text{ Volts}$$

⑥ Find V_L ... $\tilde{V}(z=0)$

$$\tilde{V}(z=0) = V_0^+ \left(e^{-j\beta(0)} + \frac{1}{3} e^{j\beta(0)} \right)$$

$$V_L = \left(7.5\angle -60^\circ \text{ V}\right)\left(1 + \frac{1}{3}\right) = 10\angle -60^\circ \text{ Volts}$$

$$10\cos(\omega t - 60^\circ) \text{ Volts}$$